

Light Rail Passenger Forecasting for Porto Metro (2012–2026)

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Forecasting Methods and Time Series

Abstract. This report forecasts monthly light rail passenger demand for Porto Metro using smoothing models, decomposition-based forecasting, SARIMA, and a SARIMAX intervention specification. The analysis uses 168 monthly observations from January 2012 to December 2025, measured in thousands of passengers. The sample begins in 2012 because the earlier 2004–2011 expansion phase does not represent the mature demand process and materially worsens residual diagnostics. Models are trained on January 2012 to December 2024, compared on the observed 2025 holdout year, and only then re-estimated for the 2026 forecast. On the 2025 holdout, the selected model is $\text{SARIMA}(3,1,0)(0,1,1,12)$, which achieves the lowest MAPE (3.66%) and the lowest MASE (0.572). The implied 2026 total is 95,479 thousand passengers, only 0.5% above the 2025 total, suggesting that the strong post-COVID recovery has largely stabilised.

Keywords: Time series forecasting · Porto Metro · Holt-Winters · STL · SARIMA

1 Introduction

The goal of this project is to forecast monthly Porto Metro passenger demand and compare competing univariate forecasting approaches before producing a final out-of-sample forecast for 2026. This is a suitable forecasting problem because the series combines a clear trend, strong seasonality at a monthly frequency, changing variance, and a large exogenous shock. Those features make it useful for comparing methods with different strengths rather than relying on a single modelling family.

The report first describes the data and the main features of the series, then applies smoothing methods, decomposition methods, and SARIMA modelling. The competing approaches are compared on a genuine holdout year, and the final forecast is produced only after model selection is complete.

2 Data, Scope and Evaluation Design

2.1 Data Source and Variable

The data are monthly light rail passenger counts for Porto Metro, measured in thousands of passengers and sourced from Statistics Portugal (INE). The avail-

Table 1. Summary of the prepared time series.

Item	Value
Start date	2012-01-01
End date	2025-12-01
Total observations	168
Mean	5,295.15
Standard deviation	1,445.88
Minimum	1,423
Maximum	9,284
Missing values after repair	0
Duplicate timestamps	0

able file originally covered January 2004 to December 2025, but the modelling sample in this report starts in January 2012.

The 2004–2011 observations were excluded deliberately rather than mechanically. During those years the Porto Metro network was still expanding, so ridership growth partly reflected system expansion rather than a stable demand process. That early-adoption phase is not representative of the mature dynamics that the 2026 forecast is meant to capture. Diagnostic checks on models with earlier starting points showed that the 2010–2011 segment inflated residual dispersion sharply and created spurious autocorrelation. Starting in 2012 therefore improves both interpretability and statistical stability.

2.2 Data Preparation

The preparation pipeline was: date parsing, chronological ordering, monthly frequency enforcement, numeric conversion, anomaly repair, and chronological splitting into training and validation periods. Only one value needed repair: a single zero observation was replaced by time interpolation. No missing values remained after cleaning.

The minimum value occurs in May 2020 and is a genuine COVID trough rather than a data-entry error. That matters because it confirms that the structural break is part of the observed demand process and not an artefact of pre-processing.

2.3 Train-Test Split

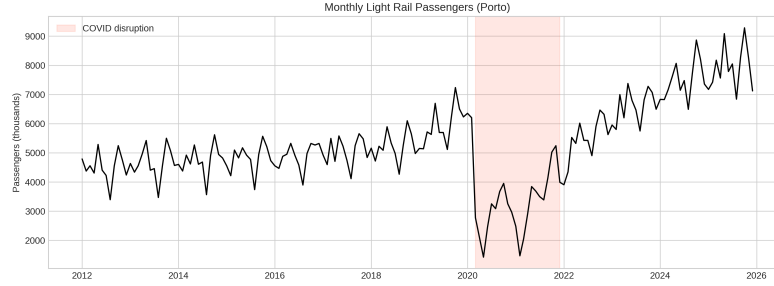
The split is fully chronological. Random splitting would be invalid because it would leak future information into model fitting. The 2025 holdout is a realistic simulation of the forecasting problem: predict one full future seasonal cycle using only earlier data.

3 Features of the Time Series

Figure ?? shows four features that drive the modelling strategy. First, the series has a clear long-run upward movement. Passenger demand in the early part

Table 2. Chronological split used in the project.

Split	Period	Observations	Role
Training	2012-01 to 2024-12	156	Model estimation
Validation	2025-01 to 2025-12	12	Model comparison
Forecast horizon	2026-01 to 2026-12	12	Out-of-sample forecast

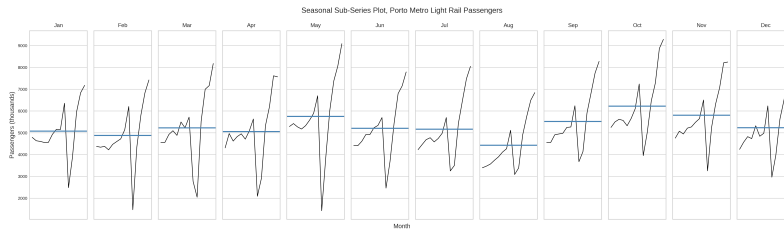
**Fig. 1.** Monthly passenger series with the COVID disruption highlighted.

of the retained sample is mostly between four and five million passengers per month, while the final years are much higher. The new sample maximum is 9,284 thousand passengers in October 2025.

Second, the series is clearly seasonal rather than being dominated by random month-to-month noise.

Figure ?? shows that the within-year pattern is strong and broadly repeated from year to year outside the COVID window. August is the persistent trough, while October is the peak and May is a secondary high point. This is consistent with urban transport demand: summer holidays reduce commuter usage in August, while October reflects the return of regular commuting and the academic calendar.

Third, the series contains a major structural break caused by COVID-19. The disruption runs from March 2020 to December 2022, covering 34 months, which is about 22% of the post-2012 modelling sample. The May 2020 trough is far below the rest of the series, and even after the first lockdown the recovery was gradual rather than immediate. This matters because standard univariate

**Fig. 2.** Seasonal subseries plot of Porto Metro demand.

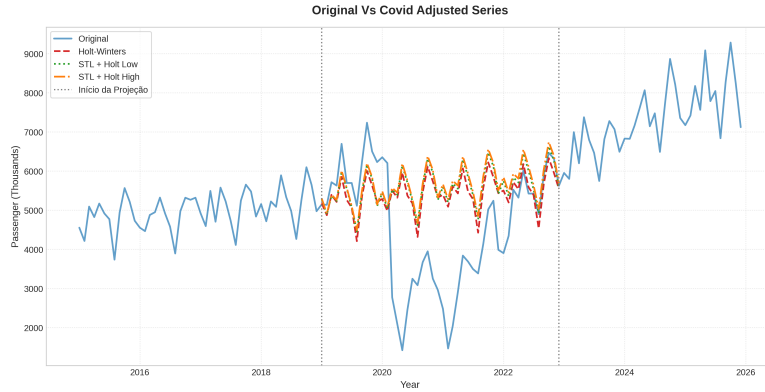


Fig. 3. Original series and the three COVID-adjusted counterfactual variants.

models assume some degree of continuity in the data-generating process; the COVID period violates that assumption.

Fourth, the series shows changing variance. Absolute month-to-month swings are larger in the high-ridership years than in the earlier years. That makes a log transformation sensible for SARIMA estimation, because modelling relative changes is more appropriate than modelling fixed absolute increments when the level itself is rising.

4 COVID Adjustment Before Smoothing

The COVID period cannot simply be ignored, but neither should it be treated as if it were a normal demand regime. Mobility restrictions were an external intervention rather than a change in the underlying seasonal demand mechanism. When Holt-Winters models are fitted directly to the raw disrupted series, the optimiser tends to collapse onto boundary solutions that freeze trend and seasonal updates. That is a sign that the model is trying to explain an exogenous shock with a structure that was never designed for it.

For that reason, three counterfactual series were built for the disruption window from March 2020 to December 2022: **series_hw**, **series_stl_low**, and **series_stl_high**. Their purpose is not to erase the shock from the historical record. Their role is narrower and more defensible: they provide a stable learning signal for models whose internal state equations would otherwise be distorted by the intervention period. Validation still uses the original observed 2025 values, so model accuracy is always assessed on real data rather than synthetic replacements.

In the final pipeline, **series_hw** is the canonical training series for the main smoothing, decomposition, and SARIMA workflow, while **series_stl_low** is also used for the multiplicative Holt-Winters specification because it produced a better-behaved seasonal scaling.

Table 3. Estimated smoothing parameters and in-sample SSE.

Method	α	β	γ	ϕ	SSE
HW_Add	0.5606	0.0000	0.0000	–	7,712,271.39
HW_Mul	0.4582	0.0000	0.0000	–	8,406,751.93
HW_Damp_Add	0.5648	0.0000	0.0000	0.9950	7,817,706.53

5 Smoothing Methods

5.1 Why Smoothing Models Make Sense Here

Smoothing models are appropriate for this dataset because monthly metro demand combines a smooth level, a persistent seasonal cycle, and relatively little short-run irregular noise outside the COVID period. They are also operationally appealing because each component has a direct interpretation.

Three Holt-Winters variants were fitted, plus a Seasonal Naive benchmark:

1. Seasonal Naive, which repeats the same month from one year earlier.
2. Holt-Winters additive (HW_Add), appropriate if the seasonal effect is roughly constant in absolute size.
3. Holt-Winters multiplicative (HW_Mul), appropriate if the seasonal amplitude scales with the level.
4. Holt-Winters damped additive (HW_Damp_Add), appropriate if the recovery trend is present but expected to flatten.

These are not arbitrary choices. The EDA suggests that both additive and multiplicative arguments are plausible: the STL seasonal factors remain similar in absolute size, but the raw series also shows larger absolute swings as the level rises. Testing both is therefore more honest than assuming one seasonal form without evidence.

5.2 Estimation Results

These estimates deserve interpretation rather than blind reporting. The level smoothing parameters are moderate, which means the fitted level responds to recent observations but does not simply copy them. That is reasonable for a monthly demand series whose underlying level changes gradually rather than abruptly.

The more important result is that all three Holt-Winters variants return $\beta = 0$ and $\gamma = 0$. These are boundary solutions. A zero β means that the trend component is set at initialisation and then never updated. A zero γ means that the seasonal factors are also fixed after initialisation. Even after COVID adjustment, the optimiser still prefers a very rigid structure.

That does not automatically invalidate the point forecasts, because a rigid seasonal template can still work well if the pattern is genuinely stable. However, it does limit what can be claimed about the model dynamics. The smoothing

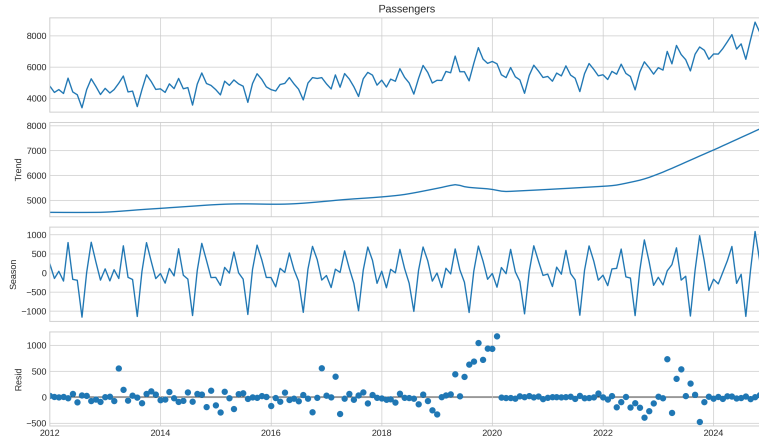


Fig. 4. STL decomposition of the training series.

models are forecasting mainly by extrapolating a fixed seasonal pattern around a level path, not by learning an actively evolving trend and seasonality.

The damped model was included because the post-COVID surge appears less steep in 2024–2025 than in 2022–2023. If growth had clearly plateaued, damping could have been beneficial. The validation results below show that this was not the case over the 2025 holdout.

6 Decomposition Methods

6.1 STL Decomposition

STL decomposition is used because it makes the structure of the series explicit. Instead of treating the data as one opaque signal, it separates the observed series into trend-cycle, seasonal, and irregular components. This is valuable here for two reasons. First, it allows the seasonal pattern to be inspected directly. Second, it reveals whether the COVID-related volatility is concentrated in the remainder rather than contaminating the long-run structure.

STL is fitted to the training series with `period = 12` and `robust = True`. The robust option matters because the transition into and out of the counterfactual COVID window can still generate edge outliers. A classical decomposition would treat those points too literally.

The decomposition supports the modelling choices made earlier. The seasonal component is clearly additive. August has the largest negative seasonal effect ($-1,087$), while October has the largest positive effect ($+783$) and May is another strong positive month ($+630$). These are stable, interpretable seasonal effects rather than random spikes.

The seasonal variation is also materially larger than the irregular variation. Across the training sample, the seasonal component has a standard deviation of about 466 thousand passengers, while the remainder has a standard deviation

of about 243 thousand. That matters because it means a large share of the predictable variation is seasonal and therefore forecastable.

The seasonally adjusted series is obtained by subtracting the seasonal component from the observed series. This transformation is analytically useful because it isolates the underlying level and trend. Appendix B reports the full decomposition from 2012 to 2025 so that the seasonal adjustment remains transparent rather than hidden inside the modelling steps.

6.2 Decomposition-Based Forecast: STL + Holt

Decomposition is not only descriptive here; it is also used as a forecasting method. The procedure is: decompose the training series with STL, remove the seasonal component to create a seasonally adjusted series, fit Holt’s linear method to that adjusted series, forecast 12 months ahead, and reconstruct the final forecast by adding back the last 12 seasonal factors.

This approach is appropriate when the seasonal pattern is stable and the main modelling challenge is the trend in the seasonally adjusted series. It is less appropriate when the recent trend slope near the forecast origin is misleading. That limitation matters here: the strong 2022–2023 recovery was still visible near the end of the training sample, so the Holt component extrapolated too much momentum into 2025. The resulting STL_ETS model underperformed the alternatives on the holdout.

7 SARIMA Modelling

7.1 Why SARIMA Is a Serious Candidate

SARIMA is appropriate for this dataset because the EDA shows persistent autocorrelation even after accounting for trend and seasonality descriptively. Unlike Holt-Winters, which treats the data through evolving components, SARIMA models the dependence structure directly. That makes it useful when short-run momentum matters in addition to the broad seasonal cycle.

7.2 Transformation and Differencing

SARIMA is estimated on the log-transformed training series. The log transformation is justified by the increasing absolute variation at higher demand levels: when the level rises, modelling relative changes is more sensible than modelling constant absolute changes.

The selected differencing scheme is $d = 1$, $D = 1$, $s = 12$. First differencing removes the long-run upward movement in the level, while seasonal differencing at lag 12 removes the repeated yearly pattern. All SARIMA forecasts and all SARIMA error metrics reported in this report are back-transformed to the original passenger scale, which makes the comparison with smoothing and decomposition models valid.

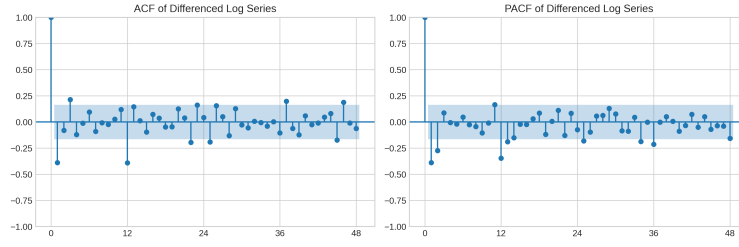


Fig. 5. ACF and PACF of the differenced log series used for SARIMA identification.

Table 4. SARIMA shortlist with information criteria and Ljung-Box diagnostics.

Model	Order	Seasonal order	AIC	AICc	BIC	LB p(12)	LB p(24)
SARIMA_310_011	(3,1,0)	(0,1,1,12)	-417.758	-417.358	-403.421	7.53e-07	6.92e-04
SARIMA_011_011	(0,1,1)	(0,1,1,12)	-414.368	-414.210	-405.789	1.20e-04	3.73e-02
SARIMA_111_011	(1,1,1)	(0,1,1,12)	-412.629	-412.364	-401.190	5.32e-05	2.55e-02
SARIMA_111_111	(1,1,1)	(1,1,1,12)	-406.002	-405.602	-391.703	4.35e-05	2.57e-02
SARIMA_011_110	(0,1,1)	(1,1,0,12)	-404.352	-404.195	-395.727	1.24e-02	3.83e-01

7.3 Shortlist Construction

The shortlist was not chosen by trial and error alone. The seasonal spike structure in the ACF suggested a seasonal moving-average component, while the short-lag ACF/PACF behaviour suggested low-order non-seasonal AR and MA terms. To avoid a narrow search, both seasonal MA and seasonal AR specifications were included.

The best information-criterion candidate is **SARIMA(3,1,0)(0,1,1,12)**. AICc is reported alongside AIC because the effective estimation sample is moderate relative to the number of parameters, so the small-sample correction is informative.

7.4 Coefficients and Interpretation

The first two autoregressive terms are strongly significant and capture short-run persistence. The seasonal MA term is also strongly significant, which is consistent with the remaining seasonal correlation structure after differencing. The third

Table 5. Coefficient estimates for the selected SARIMA model.

Parameter	Coefficient	Std. error	t-statistic	p-value	95% CI
ar.L1	-0.498	0.0639	-7.798	<0.001	[-0.623, -0.373]
ar.L2	-0.264	0.0748	-3.532	<0.001	[-0.411, -0.118]
ar.L3	0.119	0.0888	1.342	0.179	[-0.055, 0.293]
ma.S.L12	-0.729	0.1138	-6.402	<0.001	[-0.952, -0.506]
sigma2	0.00210	0.00018	11.849	<0.001	[0.00175, 0.00245]

AR term is not individually significant at conventional levels, but the model was retained because, as a complete specification, it performed best on the 2025 holdout and best on AICc among the shortlisted SARIMA candidates.

7.5 SARIMAX with a COVID Intervention Dummy

As a robustness check, a $\text{SARIMAX}(3, 1, 0)(0, 1, 1, 12)$ model was also estimated with an intervention dummy equal to 1 from March 2020 to December 2022 and 0 otherwise. Dummy (binary) variables Indicator variables (0 or 1) used to isolate anomalous events or structural shifts. A step indicator running in the above window can effectively model the structural demand shock induced by the COVID-19 pandemic, preventing the unusual variance from biasing the model's core autoregressive parameters.

The intervention model is competitive but not superior. On the 2025 holdout it records MAPE 4.11%, MAE 324.39, RMSE 395.42, and MASE 0.671. It therefore improves on the naive benchmark and on $\text{STL} + \text{Holt}$, but it is less accurate than the selected SARIMA and weaker than both Holt-Winters additive and multiplicative on MAPE. In this dataset, smoothing the disruption into a coherent training signal appears to restore the seasonal and trend structure more effectively than a single intervention dummy.

7.6 Residual Diagnostics

Residual diagnostics were examined for each shortlisted SARIMA model. The selected SARIMA and the intervention-based SARIMAX diagnostics are shown in Appendix C.

The most important result is that all shortlisted SARIMA models fail the Ljung-Box test at lag 12, and all but one still fail clearly at lag 24. This cannot be dismissed as a minor nuisance. It means that residual autocorrelation remains after modelling, so the strict white-noise assumption is violated.

In this project, the most plausible reason is the COVID structural break. The intervention created a concentrated block of unusual behaviour that standard seasonal differencing cannot fully absorb, even after the main smoothing workflow uses a counterfactual training series. In practical terms, this has two implications: the point forecasts are still meaningful and can be compared honestly on the holdout, but the nominal 95% prediction intervals should be treated as approximate rather than perfectly calibrated.

8 Forecast Evaluation on the 2025 Holdout

All methods were evaluated on the same 12 observed months of 2025. No model was trained on those values.

The evaluation metrics were ME, MAE, RMSE, MAPE, and MASE. MAPE is a defensible primary criterion in this setting because planning errors should be judged relative to the demand level, not only in absolute passengers. That

Table 6. Forecast comparison on the 2025 holdout year.

Model	ME	MAE	MSE	RMSE	MAPE (%)	MASE
Best_SARIMA	-231.60	276.44	155,990.51	394.96	3.659	0.572
HW_Mul	62.98	295.90	110,469.49	332.37	3.806	0.612
HW_Add	96.83	312.12	160,748.29	400.93	3.864	0.646
SARIMAX_Int	-13.80	324.39	156,354.46	395.42	4.107	0.671
HW_Damp_Add	157.41	346.66	179,332.33	423.48	4.267	0.717
STL_ETS	-397.78	434.93	264,951.16	514.73	5.700	0.900
SeasonalNaive	436.50	483.33	326,281.83	571.21	6.011	1.000

matters when peak months and trough months differ materially in size. It is also safe here because the validation months are all well above zero. MASE is reported alongside it to show whether each model adds value beyond the seasonal benchmark.

The validation results reveal three main points. First, every trained model beats the Seasonal Naive benchmark ($MASE < 1$), so all model-based approaches add value relative to simply repeating 2024.

Second, **Best_SARIMA** is the strongest model on the selected criteria. It has the lowest MAE, the lowest MAPE, and the lowest MASE. Its MAPE of 3.66% means that, on average, the monthly forecast deviates from the observed 2025 values by less than four percent. For a one-year-ahead transport demand forecast, that is a strong result.

Third, model choice is not completely one-sided. **HW_Mul** has the lowest RMSE (332.37), which means it avoids large isolated misses slightly better than SARIMA. The intervention-based **SARIMAX_Int** is also respectable, with RMSE almost identical to **Best_SARIMA**, but its MAPE and MAE are both worse. Once proportional accuracy and benchmark-relative accuracy are prioritised, SARIMA is the better overall choice.

The weaker performance of **STL_ETS** is also informative. Its negative mean error shows systematic over-forecasting after reconstruction, which is consistent with the earlier interpretation: the trend estimated from the post-COVID rebound was too optimistic for the more moderate 2025 pattern.

8.1 Empirical Prediction-Interval Coverage

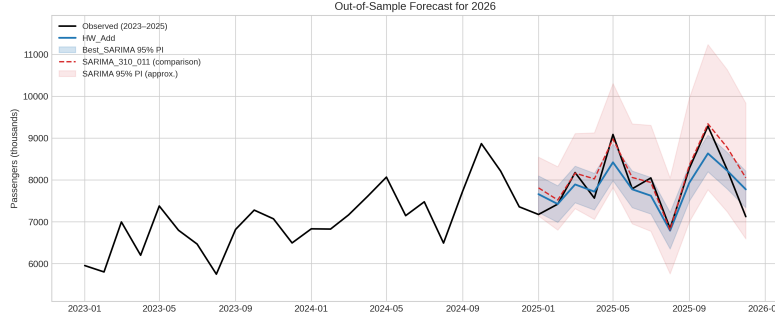
The 95% validation intervals were checked empirically where available. SARIMA covered 12 of the 12 holdout months, while **HW_Add** covered 8 of the 12 months. The SARIMA coverage looks reassuring on this holdout, but it does not remove the diagnostic caveat from the residual tests. Good empirical coverage over 12 months is useful evidence, not proof that the interval model is correctly specified.



Fig. 6. Forecast comparison on the 2025 holdout year.

Table 7. 2026 monthly forecasts from the selected SARIMA model with 95% prediction intervals.

Month	Forecast	Lower 95%	Upper 95%
Jan-2026	7,581	6,930	8,293
Feb-2026	7,380	6,672	8,162
Mar-2026	7,990	7,156	8,921
Apr-2026	7,792	6,849	8,864
May-2026	8,853	7,707	10,170
Jun-2026	7,841	6,759	9,097
Jul-2026	7,843	6,686	9,201
Aug-2026	6,699	5,660	7,928
Sep-2026	8,186	6,857	9,772
Oct-2026	9,173	7,617	11,047
Nov-2026	8,483	6,989	10,296
Dec-2026	7,658	6,261	9,368

**Fig. 7.** Out-of-sample forecast for 2026 from the selected SARIMA model.

9 Final Forecasts for 2026

The selected model is re-estimated on the full adjusted series through December 2025 and then projected 12 months ahead. Because the selected model is already the best SARIMA specification, the final forecast and the SARIMA reference forecast are the same object in the current pipeline.

The implied annual total for 2026 is 95,479 thousand passengers. The actual total for 2025 is 95,019, so the forecast implies only a 0.5% increase year on year. This is a meaningful conclusion: the model is not projecting another strong rebound. Instead, it suggests that Porto Metro is entering a more stable demand regime around a high level.

The seasonal interpretation remains consistent with the historical pattern. October is the forecast peak (9,173 thousand), while August is the trough (6,699 thousand). This matches the STL seasonal component and reinforces the view that the underlying seasonal cycle is still intact even after the COVID shock.

The prediction intervals widen as the horizon progresses, especially around the seasonal peak. That is natural for a dynamic model, but the intervals should

still be interpreted carefully because the SARIMA residuals are not white noise in the strict diagnostic sense.

10 Benefits and Limitations of the Competing Approaches

Seasonal Naive. The Seasonal Naive model is valuable as a benchmark because it is transparent and often difficult to beat when seasonality is stable. Here it underperforms because it ignores the post-COVID level recovery and simply repeats 2024. Its large positive mean error shows systematic under-forecasting of the stronger 2025 realised demand.

Holt-Winters models. The Holt-Winters models are attractive because they produce interpretable forecasts and strong validation performance. `HW_Mu1` is especially competitive and gives the best RMSE in the study. That supports the idea that some level-dependent seasonal scaling is present in the data. Their limitation is structural rigidity: the boundary solutions mean that the models are relying on fixed trend and seasonal states after initialisation. This is workable when the pattern is stable, but it weakens confidence in their adaptability if the post-COVID regime changes again.

STL + Holt. The decomposition approach adds analytical value because it makes the components explicit and yields a transparent seasonally adjusted series. It is a sensible method when the seasonal pattern is stable and the trend can be forecast separately. Its limitation in this dataset is that the trend estimate near the forecast origin still reflects the rebound phase too strongly. That made the reconstructed 2025 forecasts too optimistic.

SARIMA and SARIMAX. The selected SARIMA has the strongest validation performance on the primary criteria and captures short-run dependence that the smoothing models cannot represent directly. The SARIMAX intervention model is an important robustness check, because it keeps the original disrupted observations and models the COVID period explicitly. In this dataset, however, the intervention formulation is less accurate than the plain SARIMA estimated on the COVID-adjusted series. Both models share the same main limitation: residual autocorrelation remains, so interval estimates should be treated more cautiously than point forecasts.

Dataset-level limitations. This is still a univariate forecast. It does not use external drivers such as fares, supply changes, strikes, macroeconomic conditions, or special events. In addition, the validation evidence is based on one holdout year only. A rolling-origin evaluation would be stronger if more time were available.

11 Conclusion

The project changed substantially once the modelling sample was restricted to the mature 2012-onward period and the COVID shock was treated as an exogenous intervention rather than ordinary signal. That change improved the

coherence of the forecasting problem and made the model comparison more meaningful.

The series is dominated by strong annual seasonality, a long-run increase in demand, and a large but clearly identifiable structural break. Those features justify comparing smoothing models, decomposition-based forecasting, and SARIMA rather than relying on a single method family.

The final evidence from the 2025 holdout supports **SARIMA(3,1,0)(0,1,1,12)** as the preferred model. It gives the lowest MAPE (3.66%), the lowest MAE, and the lowest MASE (0.572). **HW_Mul** remains a credible alternative if the emphasis is on RMSE, while **SARIMAX_Int** shows that explicit intervention modelling is viable but not superior in this case.

The resulting 2026 forecast implies that Porto Metro demand remains high but broadly stabilised, with an annual total of 95,479 thousand passengers and only a modest increase over 2025. The pattern remains strongly seasonal, with the usual August trough and October peak.

The main caution is that SARIMA residuals still fail white-noise diagnostics, so the 95% prediction intervals should be read as indicative rather than exact. Even with that caveat, the point-forecast evidence is strong enough to justify SARIMA as the final forecasting model in this project.

A Monthly 2025 Validation Errors

Table 8. Monthly 2025 forecast errors for the main competing models. Error columns are forecast minus actual.

Month	Actual	Best SARIMA	Error	SARIMAX Int	Error	HW Mul	Error
2025-01	7174	7810.0	636.0	7806.4	632.4	7652.7	478.7
2025-02	7421	7520.4	99.4	7522.4	101.4	7302.6	-118.4
2025-03	8175	8160.8	-14.2	7915.8	-259.2	7926.2	-248.8
2025-04	7565	8024.6	459.6	7867.1	302.1	7705.1	140.1
2025-05	9085	8974.1	-110.9	8556.1	-528.9	8730.1	-354.9
2025-06	7788	8057.4	269.4	7856.0	68.0	7851.0	63.0
2025-07	8046	7938.4	-107.6	7746.8	-299.2	7547.2	-498.8
2025-08	6840	6803.8	-36.2	6912.5	72.5	6456.4	-383.6
2025-09	8274	8345.7	71.7	8033.8	-240.2	8003.6	-270.4
2025-10	9284	9339.4	55.4	8748.0	-536.0	9005.5	-278.5
2025-11	8244	8773.5	529.5	8338.8	94.8	8417.0	173.0
2025-12	7123	8050.2	927.2	7880.9	757.9	7665.7	542.7

B Full STL Decomposition, 2012–2025

Table 9: STL decomposition for the full 2012–2025 series.

Month	Observed	Trend	Seasonal	Remainder	Seasonally adjusted
2012-01	4788.0	4513.2	247.2	27.6	4540.8

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Month	Observed	Trend	Seasonal	Remainder	Seasonally adjusted
2012-02	4374.0	4512.2	-144.1	5.9	4518.1
2012-03	4555.0	4511.4	43.7	-0.1	4511.3
2012-04	4306.0	4510.8	-211.6	6.9	4517.6
2012-05	5286.0	4510.5	799.7	-24.1	4486.3
2012-06	4405.0	4510.4	-168.5	63.1	4573.5
2012-07	4224.0	4510.8	-195.0	-91.7	4419.0
2012-08	3392.0	4511.4	-1157.1	37.7	4549.1
2012-09	4555.0	4512.4	15.1	27.5	4539.9
2012-10	5244.0	4513.8	802.1	-71.9	4441.9
2012-11	4754.0	4515.6	276.1	-37.7	4477.9
2012-12	4239.0	4518.2	-191.6	-87.5	4430.6
2013-01	4633.0	4525.2	103.2	4.5	4529.8
2013-02	4337.0	4534.2	-209.0	11.8	4546.0
2013-03	4555.0	4545.2	87.3	-77.5	4467.7
2013-04	4963.0	4557.7	-142.8	548.1	5105.8
2013-05	5421.0	4571.2	702.5	147.3	4718.5
2013-06	4405.0	4585.5	-113.7	-66.8	4518.7
2013-07	4457.0	4600.5	-172.4	28.9	4629.4
2013-08	3468.0	4615.3	-1137.6	-9.7	4605.6
2013-09	4555.0	4629.1	41.4	-115.5	4513.6
2013-10	5498.0	4641.8	794.7	61.5	4703.3
2013-11	5072.0	4653.6	294.8	123.6	4777.2
2013-12	4566.0	4665.2	-148.0	48.8	4714.0
2014-01	4602.0	4676.9	-26.8	-48.1	4628.8
2014-02	4376.0	4688.5	-268.8	-43.7	4644.8
2014-03	4920.0	4700.4	122.0	97.6	4798.0
2014-04	4617.0	4711.9	-71.5	-23.4	4688.5
2014-05	5268.0	4722.4	615.8	-70.2	4652.2
2014-06	4608.0	4732.4	-56.7	-67.7	4664.7
2014-07	4682.0	4742.5	-160.0	99.6	4842.0
2014-08	3565.0	4752.5	-1111.9	-75.7	4676.9
2014-09	4905.0	4763.2	65.3	76.5	4839.7
2014-10	5617.0	4774.7	779.7	62.6	4837.3
2014-11	4943.0	4786.3	317.3	-160.5	4625.7
2014-12	4821.0	4797.0	-117.4	141.3	4938.4
2015-01	4554.0	4806.9	-138.0	-114.9	4692.0
2015-02	4215.0	4816.0	-328.1	-272.9	4543.1
2015-03	5093.0	4823.7	150.6	118.7	4942.4
2015-04	4825.0	4829.5	3.7	-8.3	4821.3
2015-05	5171.0	4833.9	523.1	-186.0	4647.9
2015-06	4918.0	4837.2	6.4	74.4	4911.6
2015-07	4772.0	4839.4	-157.7	90.3	4929.7
2015-08	3738.0	4840.9	-1082.1	-20.7	4820.1
2015-09	4943.0	4841.2	97.2	4.6	4845.8
2015-10	5566.0	4840.3	735.1	-9.4	4830.9
2015-11	5212.0	4839.2	330.0	42.8	4882.0
2015-12	4731.0	4838.3	-113.1	5.8	4844.1
2016-01	4554.0	4838.1	-138.6	-145.5	4692.6
2016-02	4466.0	4838.7	-366.8	-5.9	4832.8
2016-03	4879.0	4840.3	123.6	-84.9	4755.4
2016-04	4950.0	4844.0	19.3	86.7	4930.7
2016-05	5323.0	4850.7	506.7	-34.5	4816.3
2016-06	4916.0	4859.9	84.6	-28.6	4831.4
2016-07	4580.0	4871.0	-211.5	-79.4	4791.5
2016-08	3895.0	4883.9	-1028.0	39.1	4923.0
2016-09	4974.0	4898.8	102.4	-27.1	4871.6
2016-10	5320.0	4914.5	697.5	-292.0	4622.5
2016-11	5267.0	4930.6	343.6	-7.2	4923.4
2016-12	5320.0	4947.2	-190.4	563.2	5510.4
2017-01	4923.0	4963.9	-75.1	34.2	4998.1
2017-02	4595.0	4980.8	-384.3	-1.5	4979.3
2017-03	5493.0	4998.1	99.7	395.2	5393.3
2017-04	4707.0	5015.0	17.3	-325.3	4689.7
2017-05	5578.0	5030.7	574.9	-27.6	5003.1
2017-06	5227.0	5045.1	119.1	62.8	5107.9

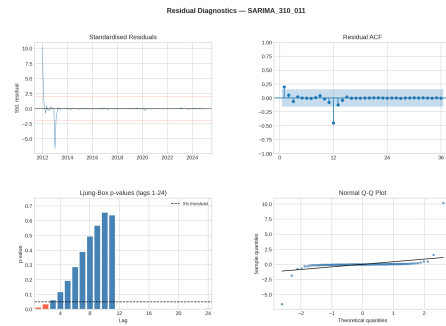
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Month	Observed	Trend	Seasonal	Remainder	Seasonally adjusted
2017-07	4738.0	5058.3	-271.1	-49.2	5009.1
2017-08	4113.0	5070.4	-989.4	32.0	5102.4
2017-09	5246.0	5082.3	67.5	96.3	5178.5
2017-10	5654.0	5094.5	678.8	-119.3	4975.2
2017-11	5479.0	5107.4	330.4	41.2	5148.6
2017-12	4839.0	5121.1	-295.2	13.2	5134.2
2018-01	5158.0	5135.7	46.9	-24.6	5111.1
2018-02	4718.0	5151.4	-384.7	-48.7	5102.7
2018-03	5224.0	5167.6	98.6	-42.2	5125.4
2018-04	5087.0	5184.5	4.8	-102.3	5082.2
2018-05	5892.0	5203.7	618.5	69.9	5273.5
2018-06	5336.0	5226.6	117.3	-7.8	5218.7
2018-07	4981.0	5254.0	-265.0	-8.1	5246.0
2018-08	4266.0	5285.0	-1002.6	-16.4	5268.6
2018-09	5264.0	5318.3	57.8	-112.2	5206.2
2018-10	6098.0	5353.2	682.3	62.5	5415.7
2018-11	5648.0	5389.1	319.8	-60.9	5328.2
2018-12	4975.0	5425.7	-269.7	-181.0	5244.7
2019-01	5149.0	5463.0	-1.7	-312.2	5150.7
2019-02	5138.0	5501.5	-374.7	11.1	5512.7
2019-03	5715.0	5542.2	129.5	43.3	5585.5
2019-04	5630.0	5585.0	-18.2	63.2	5648.2
2019-05	6696.0	5618.0	632.7	445.3	6063.3
2019-06	5696.0	5593.2	84.3	18.5	5611.7
2019-07	5696.0	5538.2	-238.6	396.3	5934.6
2019-08	5114.0	5508.0	-1030.7	636.6	6144.7
2019-09	6237.0	5493.4	44.4	699.2	6192.6
2019-10	7236.0	5481.7	706.6	1047.7	6529.4
2019-11	6502.0	5469.5	309.9	722.6	6192.1
2019-12	6231.0	5454.9	-215.4	991.5	6446.4
2020-01	6353.0	5431.1	-15.3	937.2	6368.3
2020-02	6204.0	5395.6	-367.8	1176.2	6571.8
2020-03	5501.8	5360.8	147.2	-6.1	5354.6
2020-04	5321.2	5345.7	-17.0	-7.4	5338.3
2020-05	5961.8	5353.4	624.4	-15.9	5337.4
2020-06	5372.7	5365.6	30.0	-23.0	5342.7
2020-07	5175.4	5377.6	-215.4	13.2	5390.8
2020-08	4318.7	5388.9	-1057.9	-12.3	5376.6
2020-09	5462.2	5399.4	45.4	17.5	5416.9
2020-10	6112.9	5409.2	712.8	-9.0	5400.2
2020-11	5738.4	5418.5	304.7	15.1	5433.6
2020-12	5326.4	5427.6	-161.7	60.5	5488.1
2021-01	5394.9	5436.6	-26.6	-15.1	5421.5
2021-02	5090.5	5445.7	-361.3	6.1	5451.8
2021-03	5610.2	5455.4	143.8	11.1	5466.4
2021-04	5429.6	5465.5	-43.1	7.2	5472.7
2021-05	6070.2	5476.0	684.2	-90.0	5386.0
2021-06	5481.1	5486.7	-35.4	29.7	5516.5
2021-07	5283.8	5497.7	-191.0	-22.9	5474.8
2021-08	4427.1	5509.0	-1037.0	-45.0	5464.1
2021-09	5570.6	5521.2	69.7	-20.3	5500.9
2021-10	6221.3	5534.5	702.2	-15.4	5519.1
2021-11	5846.8	5548.9	284.3	13.6	5562.5
2021-12	5434.8	5564.3	-212.6	83.0	5647.3
2022-01	5503.3	5580.7	-79.4	2.0	5582.7
2022-02	5198.9	5597.9	-358.1	-40.9	5557.0
2022-03	5718.6	5618.8	122.0	-22.3	5596.6
2022-04	5538.0	5651.9	-85.6	-28.4	5623.6
2022-05	6178.6	5696.1	808.4	-325.9	5370.2
2022-06	5589.4	5743.0	-107.0	-46.5	5696.5
2022-07	5392.2	5791.6	-148.4	-251.0	5540.6
2022-08	4535.4	5841.7	-1009.8	-296.5	5545.2
2022-09	5679.0	5892.8	82.2	-296.0	5596.8
2022-10	6329.7	5946.0	911.1	-527.4	5418.6
2022-11	5955.2	6001.5	213.7	-260.1	5741.4

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Month	Observed	Trend	Seasonal	Remainder	Seasonally adjusted
2022-12	5543.2	6058.3	-375.8	-139.4	5918.9
2023-01	5953.0	6117.1	-197.4	33.2	6150.4
2023-02	5801.0	6182.5	-353.8	-27.7	6154.8
2023-03	6995.0	6256.7	232.8	505.5	6762.2
2023-04	6200.0	6337.5	-151.7	14.2	6351.7
2023-05	7376.0	6422.5	937.4	16.1	6438.6
2023-06	6797.0	6508.5	-137.7	426.2	6934.7
2023-07	6467.0	6594.4	-75.1	-52.3	6542.1
2023-08	5748.0	6679.5	-1031.8	100.2	6779.8
2023-09	6817.0	6762.0	128.4	-73.4	6688.6
2023-10	7278.0	6839.9	1025.3	-587.2	6252.7
2023-11	7070.0	6913.4	184.9	-28.2	6885.1
2023-12	6494.0	6984.4	-587.8	97.4	7081.8
2024-01	6831.0	7055.2	-327.2	103.0	7158.2
2024-02	6825.0	7126.3	-341.7	40.5	7166.7
2024-03	7163.0	7196.6	297.2	-330.7	6865.8
2024-04	7612.0	7263.6	-209.0	557.3	7821.0
2024-05	8066.0	7325.8	1076.3	-336.1	6989.7
2024-06	7146.0	7382.7	-165.3	-71.4	7311.3
2024-07	7476.0	7435.6	3.6	36.8	7472.4
2024-08	6491.0	7485.9	-1048.9	54.0	7539.9
2024-09	7737.0	7535.1	176.3	25.7	7560.7
2024-10	8866.0	7582.6	1151.4	131.9	7714.6
2024-11	8211.0	7628.5	151.8	430.7	8059.2
2024-12	7357.0	7672.5	-791.6	476.1	8148.6
2025-01	7174.0	7714.3	-460.4	-79.9	7634.4
2025-02	7421.0	7752.8	-327.4	-4.4	7748.4
2025-03	8175.0	7790.9	363.4	20.7	7811.6
2025-04	7565.0	7828.9	-267.2	3.3	7832.2
2025-05	9085.0	7866.8	1214.8	3.4	7870.2
2025-06	7788.0	7904.7	-192.0	75.3	7980.0
2025-07	8046.0	7942.5	88.2	15.3	7957.8
2025-08	6840.0	7980.3	-1074.9	-65.4	7914.9
2025-09	8274.0	8017.9	229.6	26.4	8044.4
2025-10	9284.0	8055.5	1279.2	-50.7	8004.8
2025-11	8244.0	8093.0	121.5	29.5	8122.5
2025-12	7123.0	8130.3	-996.5	-10.8	8119.5

C Residual Diagnostics



../outputs/figures/residuals_SARIMAX_Intervention.png

Fig. 8. Residual diagnostics for the selected $\text{SARIMA}(3,1,0)(0,1,1,12)$ model (left) and the intervention-based $\text{SARIMAX}(3,1,0)(0,1,1,12)$ model (right).